

AKSHAT GUDURU

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EDUCATION

University of Central Florida - Expected Graduation: 2027

B.S. Computer Science | B.S. Statistics | Minor: Mathematics

Relevant Coursework

Calc. With Analytical Geometry II/III • Discrete Structures • Big Data Analytics • Statistical Methods I/II/III

Computer Science I/II • Algos for ML • Stats. Theory I • Numerical Calculus • Optimization • Artificial Intelligence

Clubs/Organizations

Burnett Honor College/Honors Undergrad Thesis • Knight Hacks • AI@UCF • Advanced Robotics Lab

SKILLS

Programming Languages: Python • SQL (Postgres/MySQL) • Java • JavaScript • C • C++ • HTML/CSS • R • MATLAB

Frameworks & Tools: Django • React TS/JS/Native • REST APIs • AWS EC2/Aurora • Flask • Git • Linux/Unix • NodeJs

Libraries: TensorFlow • Matplotlib • Pandas • LangChain • OpenCV • PyTorch • Gym • FastAPI • SQLAlchemy

EXPERIENCE

MARL Research Assistant — UCF

November 2024 – Present

- Collaborated with PhD researchers on decentralized **Multi-Agent RL (MARL)**, contributing to studies that examine stability and sample efficiency under limited communication and partial observability
- Implemented **PettingZoo** environments and evaluate temporal-difference update schemes using metrics like Mean Squared Bellman Error (**MSBE**) and consensus error to compare learning dynamics across settings using **Python**
- Currently leading an Honors Thesis on **fault tolerance** in MARL for robotic swarms, building a fault-injection harness (message loss, latency spikes, agent dropout) to quantify performance degradation and recovery

Machine Learning Engineer — JNTU Investment Collective

May 2025 – Present

- Explored and benchmarked **TFT** and **Autoformer** (Transformer-based) alongside **DSSM** (state-space) for multi-horizon forecasting across equities and forex, focusing on stability through regime shifts
- Curated 5 years of minute- and daily-resolution data and engineer domain features (trend, momentum, volatility, market timing indicators) with consistent scaling, windowing, and leakage-safe splits
- Developed a modular data ingestion + evaluation pipeline (**Python/SQL/ETL**) to produce standardized technical evaluations across models; summarized lifecycle impact and trade-offs (accuracy vs. compute/cost)
- Partnered with senior ML engineers to benchmark against **GRU/LSTM** stacks, introducing **Autoformer** into workflows and producing investor-ready reports using MASE, RMSE, and directional-accuracy analyses

PROJECTS

Trading Algorithm

July-August 2025

- Engineered ensemble pipeline combining **Random Forest**, **XGBoost**, and **LSTM** models for pattern recognition, an **ARIMA** model for forecasting accuracy, and **GARCH** volatility modeling for dynamic risk assessment and position sizing
- Build an **ETL** that ingests prices and fundamentals, computes 90+ technical indicators, and stores features in **AWS Aurora PostgreSQL** with job orchestration and data-freshness checks; Used **PCA** for dimension reduction during training
- Backtested strategies emphasizing Sharpe and drawdown discipline; implemented a **monitoring dashboard** in **React** surfacing ETL health, DB status, data visualizations, and strategy performance for quick failure diagnosis and iteration
- Currently working on proposed model based on Transformer Architecture for Time-Series based data

Automated Basketball Stat Tracker

June 2025

- Designed an embedded CV system (**Raspberry Pi + Arduino camera**) and fine-tuned **YOLOv8** in **Python** on custom-labeled footage to detect players, referees, and ball under varied lighting and motion
- Collected multiple datasets containing imaged and labels of basketballs, players, and referees and wrote an **ETL Pipeline** to preprocess and combine all into a single labeled unified dataset
- Implemented real-time event logic for rebounds, shots, turnovers, and travels; wrote a stats engine persisting player/team metrics and visualizing trends; achieve **89% accuracy** vs human benchmarks across 3 games
- Created a Stats Engine that detected actions and compiled stats by players and stored them to a **PostgreSQL DB**

Syllabi.AI — Project and Backend Lead

February-April 2025

- Lead a 7-dev team building an AI based **React/Flutter** web and mobile application that converts PDF syllabi into interactive learning artifacts (chapters, summaries, quizzes), coordinating roadmap, reviews, and rapid iteration
- Designed the backend (**NodeJs, MongoDB, AWS EC2**) and integrate **GPT-4o** with a Pinecone **RAG** system to answer syllabus-specific queries with source-linked responses; Implemented CI/CD pipeline for automated deployment to EC2